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1 # 🏠 MedRAG – AI Medical Assistant (RAG-based LLM System)
2
3 > **An end-to-end Retrieval-Augmented Generation (RAG) system that empowers healthcare professionals with
instant, grounded, context-aware answers from the Merck Medical Manual – 4,000+ pages of clinical knowled
ge, at your fingertips.**
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16 ## 📌 Table of Contents
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32 ## 📄 Problem Statement
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34 Healthcare professionals face a **critical information bottleneck**: thousands of pages of clinical knowl
edge, yet no fast way to extract accurate, context-aware answers.
35
36 | Challenge | Impact |
37 |---|---|
38 | Information overload from 4,000+ page manuals | Slow decision-making in critical moments |
39 | Traditional keyword search lacks medical context | Missed diagnoses, incomplete treatment plans |
40 | LLMs alone hallucinate clinical facts | Unsafe for medical settings without grounding |
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41 | No standardized knowledge retrieval tool | Inconsistent care practices across departments |
42
43 > **Real queries this system answers:**
44 > - *"What is the protocol for managing sepsis in a critical care unit?"*
45 > - *"Can appendicitis be cured via medicine, or is surgery required?"*
46 > - *"What are the diagnostic steps for suspected endocrine disorders?"*
47
48 ---
49
50 ## 💡 Solution & Architecture
51
52 This project implements a **production-grade RAG pipeline** that grounds every LLM response in verified m
53 edical text from the Merck Manual, dramatically reducing hallucinations.
54
55
56 | RAG PIPELINE |
57 | | |
58 | 📄 Merck Manual PDF (4,000+ pages) |
59 | | |
60 | ▼ |
61 | 🗂️ Text Chunking (RecursiveCharacterTextSplitter) |
62 | | |
63 | ▼ |
64 | 🧠 Embedding Generation (sentence-transformers) |
65 | | |
66 | ▼ |
67 | 📁 Vector Storage (ChromaDB) |
68 | | |
69 | ▼ |
70 | 🔍 Semantic Retrieval (Top-K relevant chunks) |
71 | | |
72 | ▼ |
73 | 🗨️ LLM Generation (Mistral-7B via llama-cpp) |
74 | | |
75 | ▼ |
76 | ✅ Grounded, Context-Aware Medical Answer |
77 | | |
78
79
80 ### Three-Stage Comparison
81
82 | Approach | Description | Quality |
83 | ---|---|---|
84 | **Baseline LLM** | Raw LLM response, no context | ⚠️ Prone to hallucination |
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85 | **Prompt-Engineered LLM** | Structured prompts, no retrieval | ◆ Better structure, still ungrounded |
86 | **RAG System** | Retrieval + Prompt Engineering + LLM | ✔ Grounded, accurate, reliable |
87
88 ---
89
90 ## 🛠️ Tech Stack
91
92 | Layer | Technology | Purpose |
93 |---|---|---|
94 | **LLM** | Mistral-7B (GGUF via llama-cpp) | Text generation |
95 | **Embeddings** | `sentence-transformers` (`all-MiniLM-L6-v2`) | Semantic vector representation |
96 | **Vector DB** | ChromaDB | Efficient similarity search |
97 | **Orchestration** | LangChain 0.3.27 | Pipeline management |
98 | **PDF Parsing** | PyMuPDF | Document ingestion |
99 | **Tokenization** | tiktoken | Token counting & management |
100 | **Runtime** | Google Colab (GPU: T4) | Accelerated inference |
101 | **Language** | Python 3.10+ | Core implementation |
102
103 ---
104
105 ## ⚡ Key Features
106
107 - 🔍 **Semantic Search** – Finds clinically relevant passages even with varied phrasing
108 - 🗂️ **Context Grounding** – Every answer is sourced from verified Merck Manual content
109 - 📊 **LLM-as-a-Judge Evaluation** – Automated quality scoring using groundedness + relevance metrics
110 - ⚙️ **Configurable Retrieval** – Adjustable top-k, chunk size, and overlap parameters
111 - 🧩 **Modular Design** – Ingestion, retrieval, and generation are fully decoupled
112 - 📺 **Prompt Engineering Baseline** – Side-by-side comparison of all three approaches
113 - 🚀 **GPU-Accelerated** – CUDA-enabled llama-cpp for fast inference
114
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116
117 ## 📁 Project Structure
118
119 ```
120 rag-medical-assistant/
121 |
122 |─ 📁 src/
123 |   |─ ingestion.py           # PDF loading & chunking logic
124 |   |─ embeddings.py         # Embedding generation & management
125 |   |─ retriever.py          # ChromaDB vector store & retrieval
126 |   |─ generator.py          # LLM loading & response generation
127 |   |─ rag_pipeline.py       # End-to-end RAG orchestration
128 |   |─ evaluator.py          # LLM-as-a-judge evaluation module
129 |
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130 |─ 📁 notebooks/
131 |   └─ MedRAG_Full_Pipeline.ipynb    # Complete experiment notebook
132 |
133 |─ 📁 data/
134 |   └─ README.md                    # Data sourcing instructions (Merck Manual)
135 |
136 |─ 📁 embeddings/
137 |   └─ .gitkeep                     # ChromaDB persistence directory
138 |
139 |─ 📁 tests/
140 |   └─ test_ingestion.py            # Unit tests for data pipeline
141 |   └─ test_retriever.py            # Unit tests for retrieval
142 |   └─ test_generator.py            # Unit tests for generation
143 |
144 |─ 📁 assets/
145 |   └─ architecture.png             # System architecture diagram
146 |
147 |─ app.py                           # (Future) Streamlit UI entrypoint
148 |─ requirements.txt                  # All dependencies pinned
149 |─ config.py                        # Centralized configuration
150 |─ .env.example                     # Environment variable template
151 |─ .gitignore                       # Git ignore rules
152 |─ README.md                        # This file
153 |``
154
155 ---
156
157 ## 🚀 Quick Start
158
159 ### Prerequisites
160
161 - Python 3.10+
162 - CUDA-compatible GPU (recommended) or CPU
163 - ~8GB RAM minimum
164
165 ### 1. Clone the Repository
166
167 ```bash
168 git clone https://github.com/YOUR_USERNAME/rag-medical-assistant.git
169 cd rag-medical-assistant
170 ```
171
172 ### 2. Install Dependencies
173
174 ```bash

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175 # GPU (recommended)
176 CMAKE_ARGS="-DLLAMA_CUBLAS=on" FORCE_CMAKE=1 pip install llama-cpp-python==0.1.85
177 pip install -r requirements.txt
178
179 # CPU fallback
180 CMAKE_ARGS="-DLLAMA_CUBLAS=off" FORCE_CMAKE=1 pip install llama-cpp-python==0.1.85
181 pip install -r requirements.txt
182 ```
183
184 ### 3. Configure Environment
185
186 ```bash
187 cp .env.example .env
188 # Edit .env with your HuggingFace token and model path
189 ```
190
191 ### 4. Add Data
192
193 ```bash
194 # Place the Merck Manual PDF in the data/ directory
195 # See data/README.md for sourcing instructions
196 ```
197
198 ### 5. Run the Pipeline
199
200 ```bash
201 # Run full RAG pipeline
202 python src/rag_pipeline.py
203
204 # Or open the notebook
205 jupyter notebook notebooks/MedRAG_Full_Pipeline.ipynb
206 ```
207
208 ---
209
210 ## 📖 How It Works
211
212 ### Step 1 – Document Ingestion
213 The Merck Manual PDF (4,000+ pages, 23 sections) is loaded using `PyMuPDFLoader` and split into semantically meaningful chunks using `RecursiveCharacterTextSplitter` with carefully tuned `chunk_size` and `chunk_overlap` to preserve clinical context across boundaries.
214
215 ### Step 2 – Embedding & Indexing
216 Each chunk is embedded using `sentence-transformers` to generate dense vector representations. These are stored in a ChromaDB persistent vector store, enabling sub-second similarity search at query time.

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217
218 ### Step 3 – Retrieval
219 At query time, the user's question is embedded and compared against all stored chunks via cosine similarity. The top-k most relevant passages are retrieved and injected into the LLM context window.
220
221 ### Step 4 – Prompt Engineering
222 A carefully designed system prompt instructs the Mistral-7B model to answer strictly from the provided context, cite limitations where information is absent, and structure responses clearly for medical professionals.
223
224 ### Step 5 – Generation
225 The Mistral-7B model (running locally via llama-cpp with CUDA) generates a grounded, coherent response using the retrieved medical context.
226
227 ---
228
229 ## 🛠️ Evaluation Methodology
230
231 This project uses LLM-as-a-Judge – a self-evaluation framework where the same LLM scores response quality on two critical axes:
232
233 | Metric | Description | What it measures |
234 |---|---|---|
235 | Groundedness | Is the answer supported by the retrieved context? | Hallucination prevention |
236 | Relevance | Does the answer actually address the user's question? | Response quality |
237
238 Both scores are generated automatically, enabling scalable, consistent evaluation without human annotation.
239
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241
242 ## 📊 Results & Key Findings
243
244 ### Qualitative Comparison (Query: Sepsis Protocol)
245
246 | Approach | Response Quality | Observations |
247 |---|---|---|
248 | Baseline LLM | Generic, vague | No specific protocols, possible hallucinations |
249 | Prompt-Engineered | Structured, clearer | Better format but still not grounded in manual |
250 | RAG System | Specific, cited | Directly references Merck Manual content |
251
252 ### Key Takeaways
253
254 - RAG significantly reduces hallucination compared to baseline LLM
255 - Prompt engineering alone improves structure but cannot substitute grounding
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256 - ChromaDB retrieval consistently surfaces relevant clinical passages
257 - LLM-as-a-Judge provides a scalable, automated quality gate
258
259 ---
260
261 ## 🧠 Future Improvements
262
263 | Enhancement | Description | Priority |
264 |---|---|---|
265 | 🔄 Hybrid Search | Combine semantic + BM25 keyword search for better recall | High |
266 | 🌐 FastAPI Backend | RESTful API for production deployment | High |
267 | 💻 Streamlit UI | Interactive chat interface for clinical staff | Medium |
268 | 🏠 HuggingFace Spaces | One-click cloud deployment | Medium |
269 | 📊 Evaluation Dashboard | Visual metrics tracking over time | Medium |
270 | 🔄 Re-ranking | Cross-encoder re-ranking for better precision | High |
271 | 📄 Multi-document | Support for multiple medical knowledge bases | Low |
272 | 🛡️ Auth & Audit | Logging & access control for clinical settings | High |
273
274 ---
275
276 ## 📁 Business Impact
277
278 This system directly addresses healthcare operational challenges:
279
280 - ⌚ **Time Savings**: Reduces information retrieval time from minutes → seconds
281 - 🎯 **Decision Support**: Grounds clinical decisions in authoritative medical literature
282 - 📋 **Standardization**: Consistent reference to the same canonical source (Merck Manual)
283 - 🗣️ **Knowledge Democratization**: Junior clinicians access senior-level knowledge instantly
284 - 🩺 **Patient Safety**: Reduced hallucination = reduced risk of erroneous recommendations
285
286 ---
287
288 ## 📄 License
289
290 This project is licensed under the MIT License. See [LICENSE](LICENSE) for details.
291
292 > ⚠️ **Medical Disclaimer**: This system is intended for research and educational purposes only. It should NOT be used as a substitute for professional medical advice, diagnosis, or treatment.
293
294 ---
295
296 ## 🙋 About
297
298 Built as an end-to-end demonstration of production-grade RAG architecture for AI/ML engineering roles. Showcases skills in LLM integration, vector databases, prompt engineering, and evaluation frameworks.
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```
299
300  **Relevant Skills Demonstrated**: RAG • LLMs • LangChain • Vector Databases • Prompt Engineering • NLP •
301  Python • ChromaDB • Sentence Transformers • LLM Evaluation
302  ---
303
304  <p align="center">★ Star this repo if you found it useful!</p>
305
```